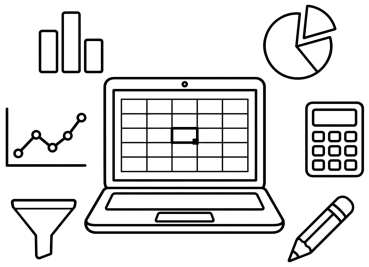
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 10 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 7
Electronic Spreadsheet



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 7 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 4, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 7 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ MS Excel 2010 or LibreOffice Calc opens on every computer.
- ☐ The projector shows a worksheet clearly from the back of the room.
- ☐ A folder is ready for each class to save the work.

"Students who type the formula themselves never forget it."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A What does it do?	6
	B Which key?	6
	C Ranges in a worksheet	8
	D Fill in the blanks	8
	E True or false	6
	F Label the Calc window	10
Chapter 2	A Work them out	10
	B Results from a worksheet	12
	C Label, value or formula?	8
	D Step by step	6
	E Write the formulas	8
	F Spot the mistakes	4
Chapter 3	A Round the numbers	6
	B Find the results	8
	C Pass or fail?	6
	D Name the tools	8
	E ICT marks of three terms	10
	F The school library	8
	G Which format?	7
Chapter 4	A How does the formula change?	8
	B Relative or absolute?	6
	C Rank the marks	4
	D Sort and filter	6
	E Which chart?	4
	F Crossword	10
End	Unit 7 Test	40
TOTAL		**223**

SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

TIP: Show the textbook screenshots on the projector and compare them with the drawings in the workbook.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (44)	Ch 2 (48)	Ch 3 (53)	Ch 4 (38)	Test (40)	Total (223)
1							
2							
3							
4							
5							
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29							

TEACHER GUIDE

Practicals at a Glance

Textbook activities on pages 195 - 243

Practical	Students do	Check that ...
1 (pp. 195 - 202)	start Excel or Calc, name the parts, the Name Box, the keys, select ranges	the Name Box shows D3; Ctrl + Home goes to A1; Ctrl selects two ranges at once.
2 (pp. 206 - 208)	the rectangle, book and pencil, and cube worksheets	C2:C4 show 138 and D2 shows 1080; new values in A2, B2 update them; D2=A2*B2 stays a label.
3 (pp. 210 - 225)	Figure 7.19 formulas, ROUND and IF, the library worksheet and its format	=SUM(B1:C2) gives 20, =AVERAGE(A1:C1,B2) gives 3.75; B6 shows 42, F3 shows 16.67.
4 (pp. 226 - 240)	the fill handle, =A2*B$2, RANK, sort, filter and the charts	C2:C7 show 10, 20, 30, 40, 50, 60; RANK of 85 is 3; the Eastern filter shows rows 6 and 8; charts have titles.

! SAFETY AND MATERIALS

- Computers with MS Excel 2010 or LibreOffice Calc. Test the program on every computer before the lesson.
- A projector for the demonstrations. Prepare Figures 7.19, 7.20 and 7.41 as files to save typing time.
- A folder for each class to save the work. No food or drinks near the computers.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - The spreadsheet and its window	194 - 202	201, 202
Chapter 2 - Data, operators and formulas	203 - 208	206, 207, 208
Chapter 3 - Functions and formatting	209 - 225	210, 216, 224
Chapter 4 - Copying formulas, sorting and charts	225 - 243	230, 237, 241 - 243
Unit 7 in One Page	244 (summary)	-

! NOTES ON THE TEXTBOOK

- Page 207, Figure 7.16: "Price of 4 books" shows =B1*3. The workbook uses **=B1*4** (it shows 192).
- Page 214: =ROUND(F3,-1) with F3 = 4.5847 gives **0**, not 10. The workbook leaves this row out.
- Page 225, step 5: the January total goes in **B6**; the average goes in B7 (step 6).
- Page 240: X runs from **-2** to 7, and the data range for the chart is **A1:K2**.
- Page 199: since version 7.4, LibreOffice Calc has 16,384 columns (not 1,024). The workbook gives only the Excel 2010 numbers.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. Which company makes Excel?	**Microsoft Corporation**
2. The cell in column D and row 3?	**D3**
3. Which keys go to cell A1?	**Ctrl + Home**
4. How many cells are in B2:C4?	**6**
5. What does every formula begin with?	**=**
6. What does =2+4*3 give?	**14**
7. Which function counts the non-empty cells?	**COUNTA**
8. What does =ROUND(456,-2) give?	**500**
9. In H$2, what does the $ keep fixed?	**the row (2)**
10. Which chart shows percentages?	**Pie chart**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Totals: 202, 162, 165, 211, 195, 167. Averages: 67.33, 54.00, 82.50, 70.33, 65.00, 55.67. Ranks: 2, 6, 5, 1, 3, 4. (Mohommad Niyas was absent in Term 2: SUM and AVERAGE use only his two marks.)

Step	The student must ...	Marks
1	Type the ICT marks worksheet (workbook exercise "ICT marks of three terms", textbook page 216).	3
2	Totals in E2:E7 with **SUM**: type it once and copy it with the fill handle.	3
3	Averages in F2:F7 with **AVERAGE**, shown with 2 decimal places.	3
4	The highest and the lowest Term 1 marks with **MAX** and **MIN**.	2
5	Ranks in G2:G7 with **RANK** and an absolute range ($E$2:$E$7).	3
6	Bold, centred headings.	2
7	A column chart of the names and the totals, with a chart title and axis titles.	3
8	Save the file with your name.	1
	Total	**20**

ANSWERS

Chapter 1

The spreadsheet and its window

A What does it do?

6 marks

1. Name Box → C	**2.** Formula Bar → E	**3.** Sheet Tab → A
4. Tabs → F	**5.** Status Bar → D	**6.** Insert Function → B

B Which key?

6 marks

1. Go to cell A1. **Ctrl + Home**	**2.** Move the active cell one cell down. **Down arrow key**
3. Go to column A of the same row. **Home**	**4.** Show the next screen of the worksheet below. **Page Down**
5. Go to the bottom right cell of the data range. **Ctrl + End**	**6.** Go to the end of the data range on the right. **Ctrl + Right arrow**

C Ranges in a worksheet

8 marks

1. B2	**2.** F2	**3.** B2:F2	**4.** H1:H7
5. R1	**6.** R2	**7.** C5:E7	**8.** 9

R1 is along a row (the row number 2 stays), R2 along a column (the column letter H stays). R3 has 3 columns x 3 rows = 9 cells. 1 mark each.

D Fill in the blanks

8 marks

1. columns	**2.** letter	**3.** number	**4.** active
5. thick	**6.** address	**7.** range	**8.** Ctrl

E True or false

6 marks

1. **T - Numbers is a spreadsheet application made by Apple Inc.**
2. **F - LibreOffice Calc is made by Microsoft Corporation.**
3. **F - In a cell address, the row number is written first.**
4. **T - The column after Z is named AA.**
5. **T - In the range A3:C3 the row number stays the same.**
6. **F - A worksheet of MS Excel 2010 has 16,384 rows.**

Statement 2: LibreOffice Calc is made by The Document Foundation. Statement 6: 1,048,576 rows; 16,384 is the number of columns.

F Label the Calc window

10 marks

1. Title Bar	**2.** Menu Bar	**3.** Standard Tool Bar	**4.** Formatting Tool Bar	**5.** Name Box
6. Formula Bar	**7.** Column Heading	**8.** Active Cell	**9.** Sheet Tab	**10.** Status Bar

1 mark each. Zoom Control and Row Heading are the two extra words.

ANSWERS

Chapter 2

Data, operators and formulas

A Work them out

10 marks

1. =6-4/2 = **4**	**2.** =5*4+3 = **23**
3. =2^5+3 = **35**	**4.** =4+5*2^3 = **44**
5. =8/4*2 = **4**	**6.** =1+2^2*3 = **13**
7. =5/2+1^3*4 = **6.5**	**8.** =3-4*3/2 = **-3**
9. =3*(2+4)/9+1 = **3**	**10.** =2^3/(5-1^3)*5 = **10**

1 mark each. ^ first, then * and / from left to right, then + and -.

B Results from a worksheet

12 marks

1. =A1*B1-D2 = **31**	**2.** =D1-C2/A2 = **-1**
3. =B3^C3-A3 = **10**	**4.** =A1*C1/C3 = **7**
5. =D2*C4*E1 = **0**	**6.** =A3/B3+A2*D1 = **9**
7. =A1+(B2*C2)-B3+C2/B2 = **23**	**8.** =B2-B3+B1*(D1+D2^B1) = **5129**
9. =A2^B3-C2+A3*(B3+C1) = **36**	**10.** =A3/C3*(D2*D1) = **12**
11. =D3/C3/B3*A3+B2 = **7**	**12.** =(D3/(A1+B2))^C2 = **1**

1 mark each. C4 and E1 are empty, so =D2*C4*E1 gives 0. In =B2-B3+B1*(D1+D2^B1) the bracket gives 2 + 4^5 = 1026 first.

C Label, value or formula?

8 marks

Cell content	Label	Value	Formula
Piumi Kaushalya	☑	☐	☐
5.72	☐	☑	☐
=A1+B1	☐	☐	☑
Kegalle	☑	☐	☐
65%	☐	☑	☐
=B1^3	☐	☐	☑
5/12/2007	☐	☑	☐
D2=A2*B2	☑	☐	☐

D2=A2*B2 does not begin with =, so it is a label. 5/12/2007 and 65% are values.

D Step by step

6 marks

=8/2*3-2^3+5 =8/2*3-8+5 =4*3-8+5 =12-8+5 =4+5 **= 9**
=5+2*3^2/(6-3) =5+2*3^2/3 =5+2*9/3 =5+18/3 =5+6 **= 11**

3 marks each: 2 marks for the steps in the right order, 1 mark for the answer (textbook Table 7.6).

E Write the formulas

8 marks

1. B3 **=B1+B2** gives **100**	**2.** B4 **=B1-B2** gives **20**
3. B5 **=B1*5** gives **300**	**4.** B6 **=B1*3+B2*4** gives **340**

1 mark for the formula, 1 mark for the result. Accept =B1*3+4*B2 and other correct orders.

F Spot the mistakes

4 marks

1. plus → **equals (=)**	**2.** right → **left**
3. addition → **multiplication**	**4.** 18 → **14**

ANSWERS

Chapter 3

Functions and formatting

A Round the numbers

6 marks

1. =ROUND(3.456,2) = **3.46**	**2.** =ROUND(7.5,0) = **8**
3. =ROUND(1234,-2) = **1200**	**4.** =ROUND(85,-1) = **90**
5. =ROUND(-2.345,1) = **-2.3**	**6.** =ROUND(0.125,2) = **0.13**

1 mark each. A 5 or more rounds up, also for negative numbers: -2.345 to 1 decimal place is -2.3 because the next digit is 4.

B Find the results

8 marks

1. =SUM(A2:C2) = **15**	**2.** =AVERAGE(A1:A3) = **4**
3. =MAX(A1:C3) = **12**	**4.** =MIN(B1:C3) = **3**
5. =COUNT(A1:F3) = **11**	**6.** =COUNTA(D1:F3) = **3**
7. =ROUND(F3,2) = **4.58**	**8.** =SUM(A1:B2)+C3 = **28**

1 mark each. COUNT(A1:F3) counts the 11 cells with numbers; COUNTA(D1:F3) counts E2, E3 and F3.

C Pass or fail?

6 marks

1. C2: **Fail**	**2.** C3: **Pass**	**3.** C4: **Fail**	**4.** C5: **Pass**

1 mark each (50 is a pass because of >=). Question 2: **=IF(B2>70,"Good","Try again")** (1 mark for the condition, 1 mark for the two words in double quotation marks).

D Name the tools

8 marks

1. Bold	**2.** Italic	**3.** Underline	**4.** Center Alignment
5. Right Alignment	**6.** Increase decimal places	**7.** Decrease decimal places	**8.** Vertical Top Alignment

E ICT marks of three terms

10 marks

1. **E2: =SUM(B2:D2) shows 202**
2. **E3: =AVERAGE(B3:D3) shows 54**
3. **B8: =MAX(B2:B7) shows 89**
4. **C8: =COUNT(C2:C7) shows 5**
5. **D8: =MIN(D2:D7) shows 60**

1 mark for each formula and 1 mark for each result. COUNT does not count "ab", so 5 students sat Term 2. Accept MAX/MIN ranges that include row 1 (the headings are text and are ignored).

F The school library

8 marks

1. **B6: =SUM(B3:B5) gives 42**
2. **B7: =AVERAGE(B3:B5) gives 14.00**
3. **E3: =SUM(B3:D3) gives 50**
4. **F3: =AVERAGE(B3:D3) gives 16.67**

1 mark for each formula and 1 mark for each result.

G Which format?

7 marks

1. the telephone number 0714365759 with its first 0: **Text**	**2.** 0.25 as 25%: **Percentage**
3. 8796093022208 as 8.80E+12: **Scientific**	**4.** 1.857142857 as 1.86: **Number**
5. 400 as Rs 400.00: **Currency**	**6.** 10:35:53 AM as 10:35: **Time**
7. 5/12/2014 as 12-May-14: **Date**	

ANSWERS

Chapter 4

Copying formulas, sorting and charts

A How does the formula change?

8 marks

1. =D2*$E2+G$2 to B1: **=E2*$E2+H$2**	**2.** =D2*$E2+G$2 to C1: **=F2*$E2+I$2**
3. =D2*$E2+G$2 to A2: **=D3*$E3+G$2**	**4.** =D2*$E2+G$2 to A3: **=D4*$E4+G$2**
5. =E2*F$2+$G2 to B1: **=F2*G$2+$G2**	**6.** =E2*F$2+$G2 to A2: **=E3*F$2+$G3**
7. =$G$2*$F2+G$2-H2 to B1: **=$G$2*$F2+H$2-I2**	**8.** =$G$2*$F2+G$2-H2 to A2: **=$G$2*$F3+G$2-H3**

1 mark each. Parts without $ move with the copy; parts with $ stay.

B Relative or absolute?

6 marks

Reference	Relative	Row absolute	Column absolute	Row and column absolute
K3	☑	☐	☐	☐
H$2	☐	☑	☐	☐
$H2	☐	☐	☑	☐
$C$1	☐	☐	☐	☑
A$1	☐	☑	☐	☐
$A1	☐	☐	☑	☐

C Rank the marks

4 marks

1. =RANK(A4,$A$2:$A$11) = **2**	**2.** =RANK(A10,$A$2:$A$11) = **4**
3. =RANK(A5,$A$2:$A$11,1) = **1**	**4.** =RANK(A8,$A$2:$A$11,1) = **3**

1 mark each. 95 is 2nd from the top; 14 is the smallest, so it is 1st in ascending order.

TEACHING TIPS

- Let pairs share a computer: one types, one checks the Formula Bar. Swap every ten minutes.
- Before Practical 2, prepare the worksheets of Figures 7.13, 7.16 and 7.17 so that slow typists can start at once.
- Say "column letter first, then row number" every time you write a cell address on the board.
- Teach the order of the operators as a chant: brackets, power, multiply and divide, add and subtract.

D Sort and filter

6 marks

1. The marks of 7.21 (A2:A11) are sorted from the smallest to the largest. What is in A2 now? **14**
2. ... and what is in A11? **96**
3. The sales table of 7.22 is filtered: Product = Laptop. Write the names that are shown. **Ashen and Amir**
4. ... and which row numbers are shown? **2, 6**
5. Now the filter is Region = Eastern. How many rows of data are shown? **2**
6. Are the hidden rows deleted? Explain. **No. Filtering only hides them for a while; Clear Filter shows them again.**

E Which chart?

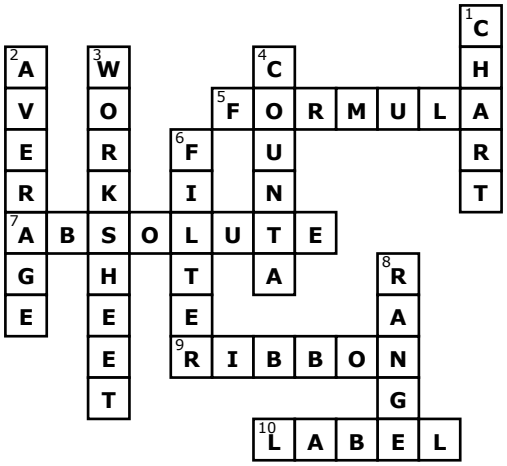
4 marks

1. Show the percentage of the total marks that each student got. **Pie chart**
2. Draw Y = X^2 - 5X - 3 from pairs of X and Y values. **XY scatter chart**
3. Show the change in electric vehicles from 2010 to 2021. **Line chart**
4. Compare the marks of six students in five subjects. **Column chart (or bar chart)**

1 mark each. For the electric vehicles a line chart shows the changes over the years best (textbook page 243, question 5).

F Crossword

10 marks



Across: 5 FORMULA, 7 ABSOLUTE, 9 RIBBON, 10 LABEL

Down: 1 CHART, 2 AVERAGE, 3 WORKSHEET, 4 COUNTA, 6 FILTER, 8 RANGE

TEACHING TIPS

- For every formula, ask two questions: what does the cell show, and what does the Formula Bar show?
- Ask students to predict a result before they press Enter. Then check it in the spreadsheet.
- Show COUNT and COUNTA side by side on a list with "ab" for absent students.

MARKING SCHEME

Unit 7 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) Microsoft Corporation	**2.** (c) 9
3. (b) 28	**4.** (b) =A5*2
5. (b) numbers	**6.** (d) $H$2
7. (c) Pie chart	**8.** (a) A, B only

2 Fill in the blanks

8 marks (1 each)

1. letter	**2.** range	**3.** handle	**4.** *
5. 9	**6.** 7.3	**7.** Cells	**8.** $ (dollar)

3 Name the parts of the window

8 marks (1 each)

1. Quick Access Tool Bar	**2.** Ribbon	**3.** Tabs	**4.** Name Box
5. Formula Bar	**6.** Active Cell	**7.** Sheet Tab	**8.** Status Bar

Accept small spelling mistakes if the part is clear (for example "Formula bar").

4 Answer these

16 marks

1. **A formula with a function for E2: the total marks of Nimal. (2 marks)**
   =SUM(B2:D2) (1 mark for SUM, 1 mark for the range). It shows 195.
2. **A formula with a function for F2: the average marks of Nimal. (2 marks)**
   =AVERAGE(B2:D2) - also accept =E2/3. It shows 65.
3. **A formula for B7: the highest ICT mark. (2 marks)**
   =MAX(B2:B6). It shows 82.
4. **A formula for H2: Pass if F2 is 50 or more, otherwise Fail. (2 marks)**
   =IF(F2>=50,"Pass","Fail") (1 mark for the condition, 1 mark for the two words in double quotation marks).
5. **A formula for G2: the rank of Nimal by the total (it is copied to G6). (2 marks)**
   =RANK(E2,$E$2:$E$6) - also accept =RANK(E2,$E$2:$E$6,0). 1 mark for RANK, 1 mark for the absolute range. Nimal is 3.

6. What does =B2+C2*2 show? Show how you work it out. (2 marks)

209 (= 65 + 144: * is done first). 1 mark for the working, 1 mark for the answer.

7. =A1*B$1 in C1 is copied to D2. Write the formula in D2. (2 marks)

=B2*C$1 (1 mark for B2, 1 mark for C$1).

8. Which chart would you use to compare the five totals? Why? (2 marks)

A column chart (or a bar chart) (1 mark): it compares classes of data items in a group (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They write the row number first (3D). A cell address is the column letter, then the row number: **D3**.
- They forget the = sign, or write **D2=A2*B2**. Nothing comes before the = sign.
- They do + before *: =2+4*3 is **14**, not 18.
- They mix the colon and the comma: A1:C1 is a range of 3 cells; A1,C1 is only 2 cells.
- They think COUNT counts everything. COUNT counts numbers only; COUNTA counts all non-empty cells.
- They forget the double quotation marks around words in IF: =IF(B2>10,"Pass","Fail").
- They forget the $ when a cell must stay fixed, or put it on the wrong part (B$2 or $B2).

HOMEWORK AND EXTRA PRACTICE

- Make a worksheet of a family's monthly spending (rice, bus fare, electricity ...) with SUM, AVERAGE, MAX and a pie chart.
- Textbook Exercise, page 241: the worksheet of the numbers 1 to 16 - questions (1) to (7).
- Textbook Activity, pages 242 - 243: the electric vehicles worksheet (an O/L style question, 1 to 8).
- Fast learners: draw the XY scatter chart of $Y = X^2 - 5X - 3$. Between which X values does the curve cross the X axis? (between -1 and 0, and between 5 and 6)

TEACHING TIPS

- Use a class marks list to teach SUM, AVERAGE, MAX, MIN and RANK. Students love to see real data.
- In IF formulas, check the double quotation marks first. A missing quote is the most common error.
- Let students type =ROUND(2.5,0) and =ROUND(2.4,0) to see how a spreadsheet rounds.
- Every chart needs a chart title and axis titles. Give a mark for each in the practical test.